Unit 10, Lesson 1: Exploring Relations

Benchmark

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

Learning Target

☐ **I can recognize relations represented in different forms.**

10.1.1 Warm-Up: Create a Pattern

1. **Place a digit in each blue box so that the numbers change by the same amount each time. Use only digits 0 to 9, at most one time each.**

□□, □□, □□, □□, ...

10.1.2 Exploration: Representing Mathematical Rules and Relations

An input-output rule is a rule that takes an input value and uses it to determine an output value.

input ⇨ rule ⇨ output

For example, in the table shown, a rule was used to get each output value from its corresponding input value.

Input	-5	$-\frac{1}{3}$	0	2.5	6
Output	-15	-1	0	7.5	18

1. **Work with your partner to determine the rule for the previous table. Then, complete the statement to describe the rule.**

Each
- ○ input
- ○ output

value is the result of taking the
- ○ input
- ○ output

and
- ○ adding 10 to it.
- ○ dividing it by 3.
- ○ multiplying it by 3.
- ○ subtracting 10 from it.

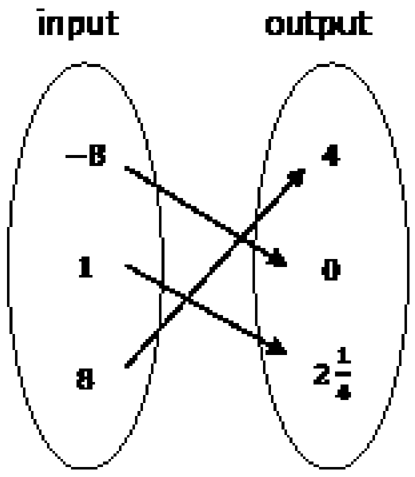
2. Work with your partner to complete the input-output table for each of the following rules. The first row has been completed as an example.

<table>
<tr><th>Rule</th><th>Table of Values</th></tr>
<tr><td>x ⇨ write 1 if the input is odd; write 0 if the input is even ⇨ y</td><td>
<table>
<tr><th>Input</th><th>Output</th></tr>
<tr><td>$\frac{1}{4}$</td><td>7</td></tr>
<tr><td>2.35</td><td></td></tr>
<tr><td>42</td><td></td></tr>
<tr><td>−1</td><td></td></tr>
</table>
</td></tr>
<tr><td>input ⇨ add 1, then multiply by 4 ⇨ output</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>−12</td><td></td></tr>
<tr><td>−5</td><td></td></tr>
<tr><td>6</td><td></td></tr>
<tr><td>19</td><td></td></tr>
</table>
</td></tr>
</table>

Each of the tables represent a *relation*.

A ***relation*** is a set of input-output pairs.

Relations can also be represented by mapping diagrams. The mapping diagram below shows input-output pairs for the rule: "divide by 4, then add 2."



3. Work with your partner to complete the statements.

In a mapping diagram an [o input / o output] **is mapped to its corresponding** [o input / o output] **using an arrow. In the previous mapping diagram above, the input 8 results in an output of** [o 4 / o 0 / o $2\frac{1}{4}$] **for the relation shown.**

4. **Work with your partner to complete the mapping diagram for each of the following rules.**

Rule	**Mapping Diagram**
input → find the square root(s) of the input → output	input: 0, 4, 16 output: −4, −2, 0, 2, 4
x → subtract 3, then multiply by $\frac{1}{2}$ → y	x: −2, 0, 5 y:

5. **Complete the statements.**

The
- o input values
- o output values

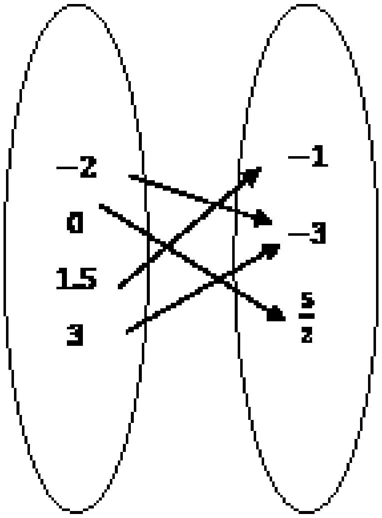
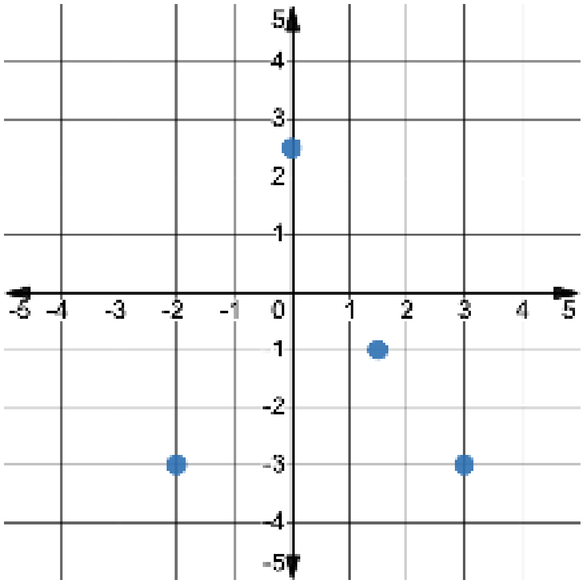
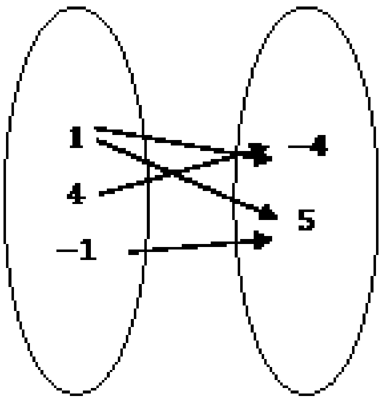
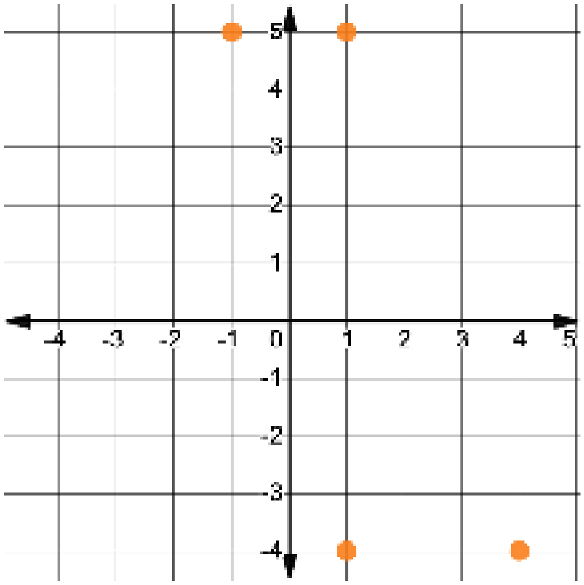
of a relation depend on the values used for the
- o inputs.
- o outputs

Therefore, the inputs are the
- o dependent variables,
- o independent variables,

and the outputs are the
- o dependent variables.
- o independent variables.

Relations can be represented using graphs.

For each relation below, its graph and mapping diagram are shown.

Relation	Mapping Diagram	Graph
Relation A		
Relation B		

6. **With your partner, discuss your observations when comparing the mapping diagrams and graphs for each relation.**

7. Complete the statements.

The mapping diagram for each of the relations has

- o 2
- o 3
- o 4

input-output pairs indicated by arrows. The graph showing the same relation has

- o 2
- o 3
- o 4

points. The input value from the mapping diagram matches the

- o x-value
- o y-value

of each point on the graph, and its corresponding output value from the mapping diagram matches the

- o x-value
- o y-value

of each point on the graph.

A set of ordered pairs can also be used to represent a relation.

8. The ordered pairs shown below can be used to represent Relation A from the previous table.

$$\{(-2, -3), \left(0, \frac{5}{2}\right), (1.5, -1), (3, -3)\}$$

Write the set of ordered pairs that represents Relation B from the previous table.

10.1.3 Bring it Together: Representing Relations

A relation is any set of input-output pairs. Relations can be represented using tables of values, mapping diagrams, a set of ordered pairs, or graphs.

1. Complete the statements.

In a relation, the input values are the

- o dependent
- o independent

variables. The input value is the

- o x-value
- o y-value

of an ordered pair. The output values of a relation are the

- o dependent
- o independent

variables. The output value is the

- o x-value
- o y-value

of the relation. The relation can be represented as a set of ordered pairs or on a graph.

10.1.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding for today's lesson.